Richard Tsai

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Highly ambitious software development engineer with 2 years of experience in full-stack development, with a focus on cybersecurity & DevSecOps. Experience with wide range of tech stacks while maintaining good cybersecurity posture & introducing Quality of Life automation with CI/CD pipelines.

Experience

Cloud Application Developer **07/2022 – Current**

IBM

- Working as a full-stack developer in a project with Gatsby, Apollo Client for GraphQL, Java Spring Boot and Jahia CMS
- Creating Java Maven project-structured Jahia Modules to support frontend content management logic
- Ensuring correct data formatting from GraphQL node hosted on Azure App Service to Gatsby App
- Utilizing Jira and BitBucket for project issues, bug-tracking, and version control
- Streamlining manual development environment setup by over 50% by implementing Docker containers and Bash scripting automation
- Building automated CI/CD pipelines on Jenkins for containerized applications using Docker & Terraform Secrets Manager
- Speeding Gatsby build time and website performance by 25% by reusing Gatsby cache in CD pipelines and optimizing business logic

Cybersecurity Defensive Operations Programming Intern **05/2021 – 08/2021**

The MITRE Corporation

- Contributed as a full-stack developer in IDS/IPS in-house project with PostgreSQL and Redis, Python Flask, and VueJS
- Utilized Jira for task tracking, assignments, and collaboration with team members
- Created new Python Flask API calls to manipulate data and fetch data from PostgreSQL tables
- Ensured accuracy and reliability of API endpoints through testing using the Python Pytest framework, contributing to CI pipeline

Cybersecurity Defensive Operations Intern **06/2020 – 08/2020**

The MITRE Corporation

- Analyzed network and host-based log data using Zeek and ELK dashboard for insights
- Developed a master bash script to automate various Zeek scripts for comprehensive network packet analysis
- Monitored and managed a web crawling application utilizing in-house Python library packages
- Created a custom admin dashboard and reporting system using AngularJS

Helpdesk Student Consultant **07/2019 – 05/2022**

Rensselaer Polytechnic Institute

- Troubleshooted printer problems throughout campus, performed LMS maintenance, and diagnosed student laptop problems
- Provided support to students & faculties via phone call and remote desktop via Bomgar
- Monitored incoming issues service tickets, track issues, and troubleshooting issues via Zendesk software

Education | Certifications

Rensselaer Polytechnic Institute **07/2018 – 05/2022**

Bachelor of Science in Computer Science

CompTIA Security+ ce: CompTIA License SY0-501 **08/2019 – 08/2025**

AWS Certified Security: SCS-C02 **11/2023 – 11/2026**

Personal Projects

YACS – Open-Source Schedule Management @ RPI RCOS

Yet Another Course Scheduler – an open-source project to help students plan out their semester

- Led a full-stack project utilizing PostgreSQL, Python FastAPI, and VueJS, containerized with Docker
- Conducted weekly agile sprint meetings, used GitHub Projects to monitor ticket progress, and issued release candidates
- Implemented NGINX best practices to enhance security and maintain optimal performance. Conducted CI testing.
- Automated in-house data migration with shell scripts for PostgreSQL tables, ensuring table-upgradability and futureproofing
- Managed GitHub Action CI/CD pipelines and deployment via Kubernetes, ensuring high availability
- Enhanced security on staging servers on OCI, AWS, and on-premises production servers using Terraform

Personal Home Lab

On-going personal home lab to add more future projects/servers

- Proficient in monitoring and securing network traffic using FortiGate NGFW with custom IPv4 policies, DoS policies, and VPN
- Deploy and manage multiple on-home services across servers and containers via Kubernetes clusters
- Set up Grafana dashboards and Prometheus data-gathering solutions to monitor server activities and network traffic
- Implemented CrowdSec open-source system as the home network's Intrusion Prevention System (IPS) to safeguard public-facing services
- Manage internal DNS servers with High-Availability via Unbound for DNSSEC validation and serve as a trust anchor
- Automate security updates for all services using Bash scripting and cron jobs

Skills

- *Programming Languages:* C, C++, Java, Python, SQL, Bash, Node.JS frameworks
- *Operating Systems / Platforms:* Windows, macOS, Linux, Amazon Web Service, Docker, Kubernetes, GitHub, GitLab, BitBucket